

N M P R O J E C T R E P O R T

AI POWERED LEGAL DOCUMENT GENERATOR

Automating the drafting of legal documents using Artificial Intelligence

Student Name

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Project Guide

Academic Year

Submitted as part of the NM Project

Acknowledgement & Declaration

Acknowledgement

I sincerely thank everyone who supported me in completing this project on the **AI Powered Legal Document Generator**.

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Declaration

I hereby declare that this project report is the result of my own work. It has not been submitted to any other institution or programme for any other purpose. All sources of information used in this report have been acknowledged.

Signature: _____

Date: _____

Abstract

Preparing legal documents such as rental agreements, affidavits and non-disclosure agreements is slow, costly and confusing for most people. Small businesses, students and first-time users often depend on outdated templates or expensive professional help.

This project presents an **AI Powered Legal Document Generator**, a web application that asks users simple questions and automatically drafts a complete, well-structured legal document. It combines a Large Language Model (LLM) with a verified clause library and a validation step, so the output is accurate, consistent and easy to understand.

The system reduces drafting time from hours to minutes, lowers cost, and makes basic legal documentation accessible to everyone. It is a drafting aid; final review by a qualified lawyer is recommended.

Keywords: Artificial Intelligence, Large Language Model, Legal Technology, Document Automation, Natural Language Processing

Project at a Glance

Item	Details
Domain	Legal Technology (LegalTech) and Artificial Intelligence
Approach	LLM drafting with clause retrieval (RAG) and rule-based validation
Input	Simple form answers given by the user
Output	Ready-to-review legal document in DOCX and PDF
Target Users	Individuals, students, startups, small businesses

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1. Introduction

Legal documents are part of everyday life: renting a house, hiring an employee, starting a business or signing an agreement. Yet drafting them requires legal knowledge, time and money that many people do not have.

Recent progress in Artificial Intelligence, especially **Large Language Models (LLMs)**, allows computers to understand instructions and produce well-structured text. This project applies that capability to legal drafting: the user answers a few questions and the system generates a complete document in minutes.

Background

- Legal drafting is repetitive: most agreements follow standard structures.
- Errors or missing clauses can cause disputes and financial loss.
- Digital India initiatives are increasing demand for online legal services.

Why AI for Legal Documents?

Benefit	Explanation
Speed	Draft ready in minutes instead of hours or days
Cost	Reduces dependence on paid drafting for routine documents
Consistency	Same quality and structure every time
Accessibility	Simple language, available 24/7 to anyone online

Important Note

This system assists with drafting. It does not replace a licensed lawyer, and generated documents should be reviewed before use.

2. Problem Statement

Ordinary users face many obstacles when they need a legal document. Professional drafting is expensive, online templates are generic, and legal language is hard to understand. As a result, people either skip proper documentation or sign documents they do not fully understand.

Key Problems

Problem	Impact on the User
High lawyer fees	Routine documents become unaffordable for individuals and startups
Time-consuming process	Days of waiting for a simple agreement
Complex legal language	Users sign terms they do not understand
Human error	Missing clauses, wrong names or dates, weak protection
Inconsistent formats	Different structure and quality across documents
Limited local access	Legal help is scarce in small towns and regional-language users

Problem in One Line

There is no simple, affordable and reliable way for common users to generate accurate legal documents quickly.

3. Objectives and Scope

Objectives

- Build a web application that generates legal documents from simple user inputs.
- Use AI to draft clear, correct and well-structured clauses.
- Validate every draft against a checklist of mandatory clauses.
- Reduce drafting time and cost for routine legal documents.
- Export the final document in editable (DOCX) and print-ready (PDF) formats.

Scope of the Project

In Scope	Out of Scope
Common documents: rental agreement, NDA, affidavit, offer letter	Court filings and litigation strategy
Guided question form and AI drafting	Providing binding legal advice
Clause validation and plain-language summary	Complex, high-value corporate contracts
DOCX and PDF export	Physical stamping and registration
English language (Tamil and Hindi planned later)	Representation before authorities

Target Users

Individuals, tenants and landlords, students, freelancers, startups and small businesses.

4. Existing Systems and Gap Analysis

Before designing the system, current ways of creating legal documents were compared.

Approach	Strengths	Limitations
Lawyer drafting	Accurate, personalised	Costly and slow
Static template websites	Cheap, quick	Rigid, generic, no customisation
General AI chatbots	Flexible, fast	May invent clauses, no validation
Proposed system	Fast, guided, validated, customisable	Needs human review for complex cases

Gap Identified

- Templates lack flexibility; chatbots lack reliability.
- Few tools combine AI drafting with a verified clause library.
- Most tools offer no automatic check for missing or risky clauses.
- Very few options support regional legal context and local languages.

Our Approach

Combine the flexibility of AI with the reliability of verified templates and automatic validation.

5. Proposed System

The proposed **AI Powered Legal Document Generator** is a web platform in which the user selects a document type, answers guided questions, and receives a complete draft. The AI engine writes the clauses, a validation module checks them, and the user can review, edit and download the final document.

Key Features

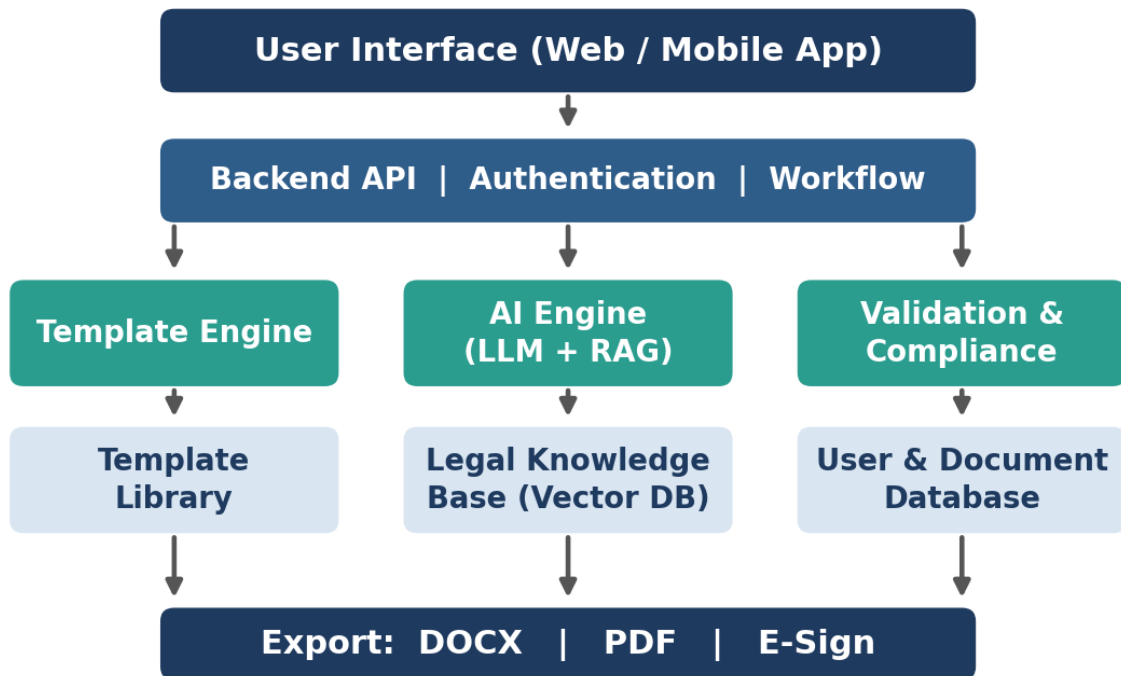
- **Guided questionnaire:** simple questions instead of legal jargon.
- **AI drafting:** clear clauses generated from user inputs.
- **Clause library:** verified standard clauses reduce errors.
- **Validation:** automatic check for missing or conflicting terms.
- **Plain-language summary:** explains the document in simple words.
- **Export:** download as DOCX or PDF.

Who Uses the System

User	How They Use It
General user	Generates documents such as rental agreements and affidavits
Small business	Creates NDAs, offer letters and service agreements
Legal reviewer	Reviews, edits and approves drafts
Administrator	Manages templates, clause library and users

6. System Architecture

The system follows a layered architecture. Each layer has a single responsibility, which keeps the system easy to maintain and extend.



- **Presentation layer:** web interface for forms, preview and download.
- **Application layer:** backend API for authentication and workflow control.
- **AI layer:** template engine, LLM with retrieval, and validation.
- **Data layer:** template library, legal knowledge base and user records.

7. Technology Stack

The technologies below are chosen for reliability, community support and ease of development.

Layer	Technology	Purpose
Frontend	React.js, Tailwind CSS	Responsive user interface
Backend	Python, FastAPI	REST APIs and business logic
AI Model	LLM API (e.g., Claude)	Clause and document generation
Retrieval	Vector database (FAISS / ChromaDB)	Find relevant clauses and legal context
Templates	Jinja2, python-docx	Fill and format documents
Database	PostgreSQL	Users, documents, audit logs
Export	python-docx, ReportLab	Create DOCX and PDF files
Security	JWT, HTTPS/TLS	Login protection and secure transfer
Deployment	Docker, cloud hosting	Portable, scalable deployment

Flexibility

Any modern LLM can be used. The architecture keeps the AI model replaceable without changing the rest of the system.

8. Key Modules

The system is divided into six modules. Each performs one clear task.

Module	Description	Output
1. User Input	Collects details through a guided form with validation	Structured data
2. Template Selection	Chooses document type and jurisdiction rules	Base template
3. AI Drafting Engine	Writes clauses from inputs using the LLM	Draft document
4. Clause Retrieval (RAG)	Fetches verified clauses and legal context to guide the AI	Reliable clauses
5. Validation & Review	Checks mandatory clauses, names, dates and conflicts	Issue report
6. Export & Storage	Creates DOCX/PDF and saves history securely	Final file

Design Principle

Separate modules make the system easy to test, upgrade and maintain.

9. Workflow

The user moves through six simple steps, from selecting a document to downloading the final file.



Step	Action	Result
1	User selects the document type	Correct template loaded
2	User answers guided questions	Inputs collected
3	AI drafts clauses using inputs and retrieved clauses	First draft
4	Validation module checks the draft	Issues flagged
5	User reviews and edits the draft	Approved version
6	System exports DOCX or PDF	Final document

10. Supported Legal Documents

The first version supports common, high-demand documents with well-defined structures.

Document	Purpose	Key Inputs
Rental / Lease Agreement	Property rent between landlord and tenant	Parties, rent, deposit, duration
Non-Disclosure Agreement	Protect confidential information	Parties, scope, term
Employment Offer Letter	Terms of a job offer	Role, salary, joining date
Affidavit	Sworn statement of facts	Name, address, statement
Power of Attorney	Authorise someone to act for you	Principal, agent, powers
Sale Agreement	Sale of goods or property	Seller, buyer, price, terms
Legal Notice	Formal notice for a dispute	Sender, receiver, issue
Service Agreement	Freelance or business services	Scope, fees, timeline

11. Implementation Approach

Prompt Design

A structured prompt controls the AI. It states the document type, jurisdiction, user inputs and strict rules, so the output stays accurate and relevant.

```
You are a legal drafting assistant.  
Document : Rental Agreement  
Region   : Tamil Nadu, India  
Inputs   : landlord, tenant, rent, deposit, duration  
Rules    : Use plain English and numbered clauses.  
          Include mandatory clauses from the library.  
          Do not invent facts. Mark missing data as [TBD].  
Output   : Structured clauses (JSON)
```

Core Logic

```
inputs  = collect_user_inputs(doc_type)  
clauses = retrieve_clauses(doc_type, region)  
draft   = llm.generate(build_prompt(inputs, clauses))  
issues  = validate(draft, mandatory_checklist)  
if issues: draft = llm.revise(draft, issues)  
file    = render_template(draft, format='DOCX')
```

Reliability

Retrieval-Augmented Generation (RAG) grounds the AI in verified clauses, which reduces made-up content.

12. Security, Privacy and Ethics

Legal documents contain personal and sensitive data, so protection is built into the design.

Area	Measure Taken
Data in transit	All traffic secured with HTTPS / TLS
Data at rest	Encrypted storage for documents and user data
Access control	Login with JWT tokens and role-based permissions
Privacy compliance	Consent, minimal data collection and deletion on request, following the principles of India's Digital Personal Data Protection Act, 2023
AI reliability	Clause retrieval, validation checks and human review to limit incorrect output
Transparency	Clear disclaimer that the tool is not a substitute for legal advice
Audit	Activity logs to trace document creation and changes

Ethical Use

The system supports users but does not give legal advice. Users are always shown a disclaimer and encouraged to consult a lawyer for important matters.

13. Testing and Evaluation

The system is tested at functional, quality and usability levels.

Sample Test Cases

Test Case	Input	Expected Result
Valid rental inputs	All fields filled correctly	Complete agreement generated
Missing field	Rent amount left blank	Form asks user to complete the field
Mandatory clause check	Draft without termination clause	Validation flags the missing clause
Export	Approved draft	Correct DOCX and PDF downloaded
Unauthorised access	Open another user's document	Access denied

Evaluation Metrics

Metric	How It Is Measured	Target
Clause completeness	Mandatory clauses present in the draft	90% or more
Generation time	Time from submit to draft	Under 30 seconds
Readability	Feedback from test users	Clear to non-lawyers
User satisfaction	Short survey after use	Positive rating

14. Benefits and Limitations

Benefits	Limitations
Saves time: drafts in minutes	Cannot replace a qualified lawyer
Reduces cost for routine documents	AI output may need correction
Simple language and guided process	Limited to supported document types
Consistent and validated structure	Laws change, so templates need regular updates
Available anytime, from anywhere	Requires internet access
Easy to extend with new templates	Regional language support still in progress

Balanced View

The tool is best for routine, standard documents. Complex or high-value matters should always be reviewed by a professional.

15. Future Enhancements

The project can grow into a complete legal assistant platform.

Enhancement	Description
Multilingual support	Generate and explain documents in Tamil, Hindi and other languages
Voice input	Let users speak their details instead of typing
E-signature and e-stamping	Sign and stamp documents digitally
Contract risk analysis	Upload an existing contract and highlight risky clauses
Lawyer review marketplace	Connect users to verified lawyers for a quick review
Mobile app	Android and iOS apps for easier access
Smart clause suggestions	Recommend extra clauses based on the user's situation

16. Conclusion

The **AI Powered Legal Document Generator** shows how Artificial Intelligence can make legal documentation faster, cheaper and easier to understand. By combining an LLM with verified clauses and automatic validation, the system produces reliable drafts while keeping the user in control.

The project achieves its main goal: helping common users create routine legal documents without deep legal knowledge. With continued improvement in language support, validation and integrations, it can become a practical tool for individuals and small businesses.

Key Takeaway

AI does not replace legal expertise. It removes repetitive drafting work so that people can get accurate, understandable documents quickly.

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